



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SEMESTER 2

2025/2026

SECP3843-02 SPECIAL TOPIC IN DATA ENGINEERING

TUTORIAL MICROSOFT AZURE

MADE BY:

NAME	MATRIC NUMBER
DAYANG FARAH FARZANA BINTI ABANG IDHAM	A23CS0071
FARRA NURZAHIN BINTI ZAHARIL ANUAR	A23CS0079
SAFIYA NURSYAHADAH BINTI MASNOOR	A23CS0176

LECTURER: DR. ARYATI BINTI BAKRI

TABLE OF CONTENT

1.0 Project Overview.....	3
2.0 Description of the Solution.....	4
3.0 Technology Use.....	9
4.0 Data Pipeline Architecture.....	10
6.0 REFLECTION.....	12
Dayang.....	12
Farra.....	12
Safiya.....	12

1.0 Project Overview

This project involves building an end-to-end data engineering pipeline on Microsoft Azure. The goal is to automate the full journey of data, from extraction at the source all the way through to a live dashboard that stakeholders can interact with. The pipeline was built by following a hands-on tutorial that uses real Azure services including Data Factory, Data Lake Gen2, Databricks, Synapse Analytics, and Power BI.

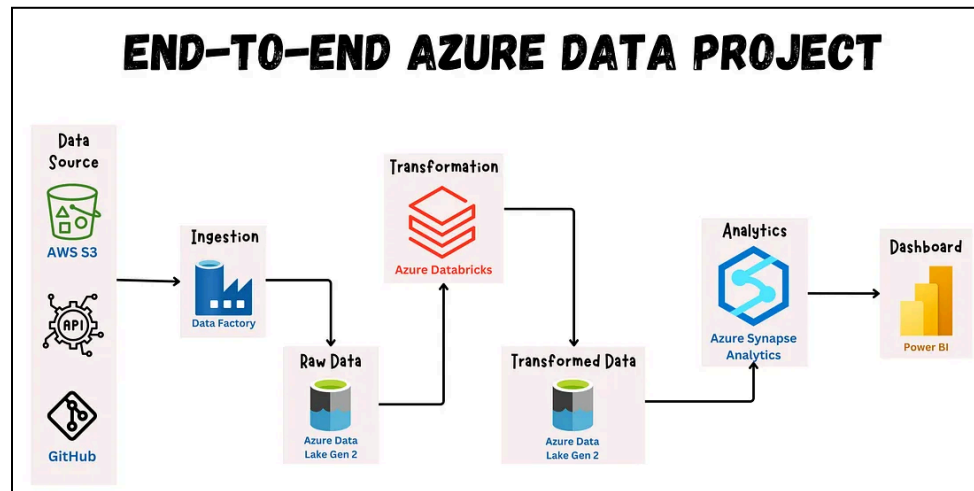


Figure 1.1: End-to-end Data Engineering Pipeline

The business scenario behind the project is based on a retail organization that stores customer and sales transaction records in an on-premises SQL Server database. The problem is that this data sits in isolation and cannot be easily analyzed or reported on in real time. By migrating the pipeline to Azure, the organization can automate data collection, apply transformations in the cloud, and present insights through a dashboard without any manual effort.

The main objectives of the project are:

- To extract data from an on-premises SQL Server database into cloud storage.
- To transform and clean the data using Azure Databricks across multiple processing layers.
- To store structured data in Azure Synapse Analytics for SQL-based querying.
- To build a Power BI dashboard that visualizes key metrics such as sales by product category and gender distribution.

The scope of the project covers the full pipeline from source to dashboard, using only free-tier Azure services where possible. Figure 1.1 shows all resources are organized under a single Resource Group to keep management centralized.

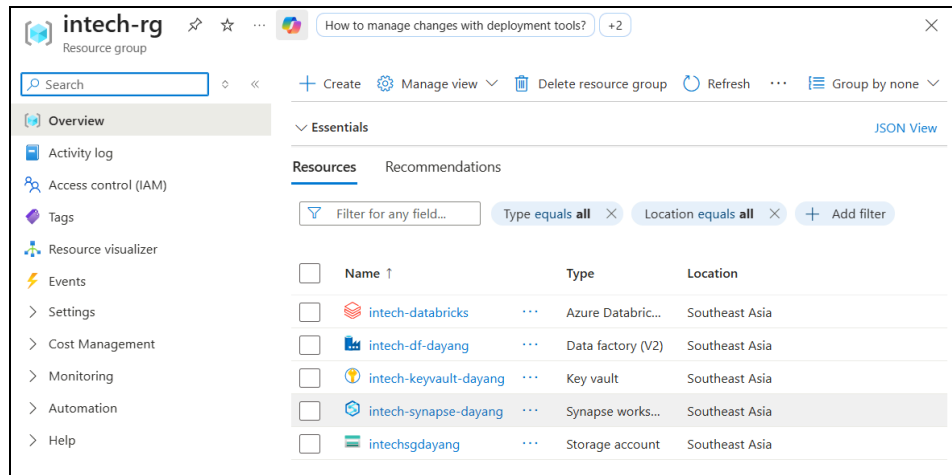


Figure 1.2: Services in Resource Group

2.0 Description of the Solution

Data ingestion is the first active stage of the pipeline and is handled entirely by Azure Data Factory (ADF). ADF acts as the orchestration engine, meaning it controls when and how data moves between systems. In this project, ADF was configured with a self-hosted integration runtime that creates a secure tunnel between the on-premises SQL Server database and the Azure cloud environment.

Figure 2.1 shows the ingestion pipeline uses a ForEach activity to loop through all the tables in the SQL database dynamically, rather than creating a separate copy activity for each table. In the same figure, it simulates how a Copy Data activity reads each table from the SQL source and writes it to the Bronze layer of Azure Data Lake Gen2 as a Parquet file. Figure 2.2 and 2.3 shows the source and parquet file in the Copy Data respectively. This approach means the pipeline automatically handles any new tables added to the source without requiring manual changes to the pipeline configuration.

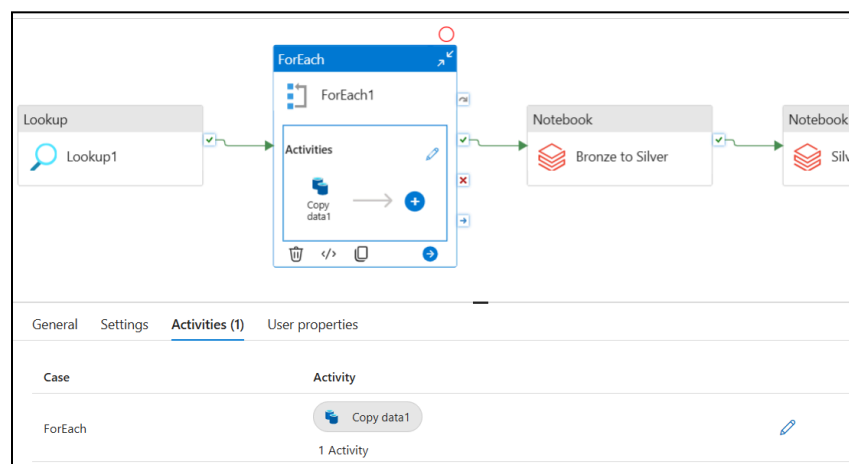


Figure 2.1: Activity in ForEach

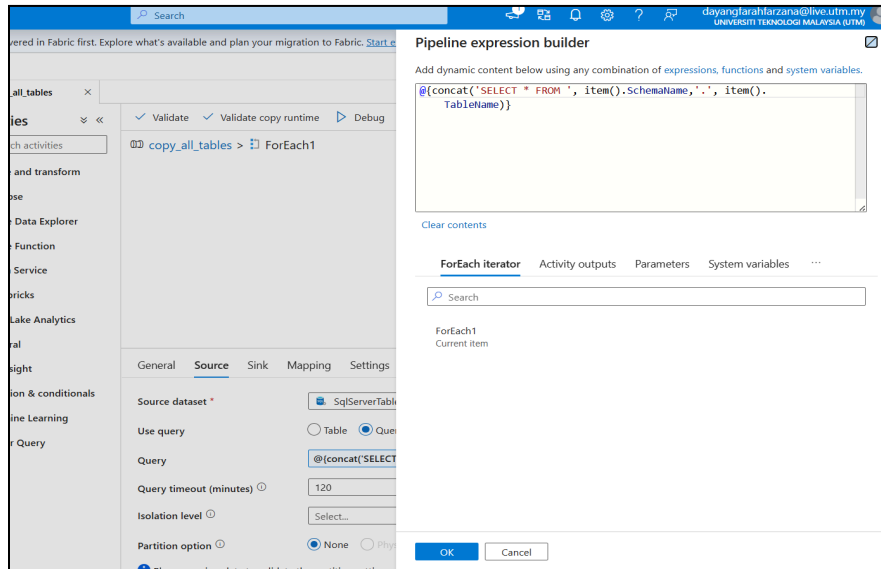


Figure 2.2: Source of Copy Data

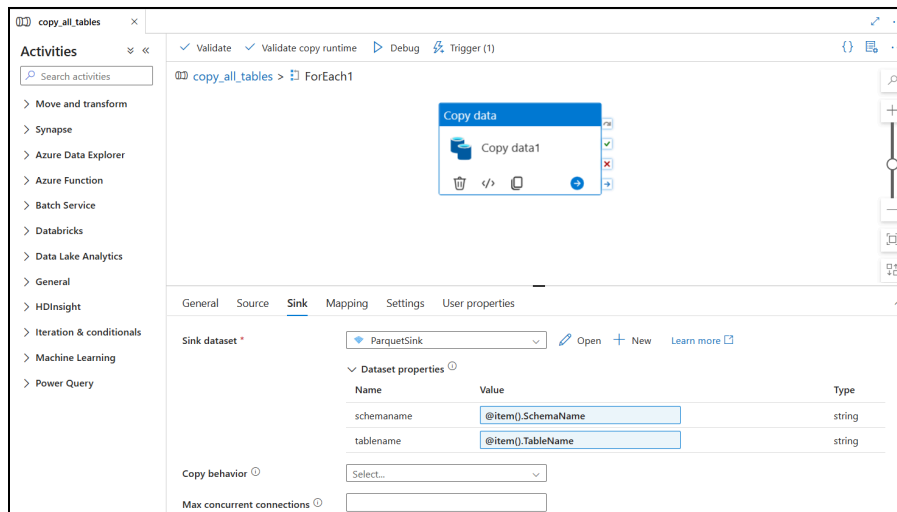


Figure 2.3: Parquet File in Copy Data

The Bronze layer in Figure 2.4 is created for raw, unmodified data lands in Storage Accounts. Nothing is changed at this stage as it is essentially a direct copy of the source data preserved in its original form. This is important because it gives the team a full audit trail of what the original data looked like before any transformations were applied.

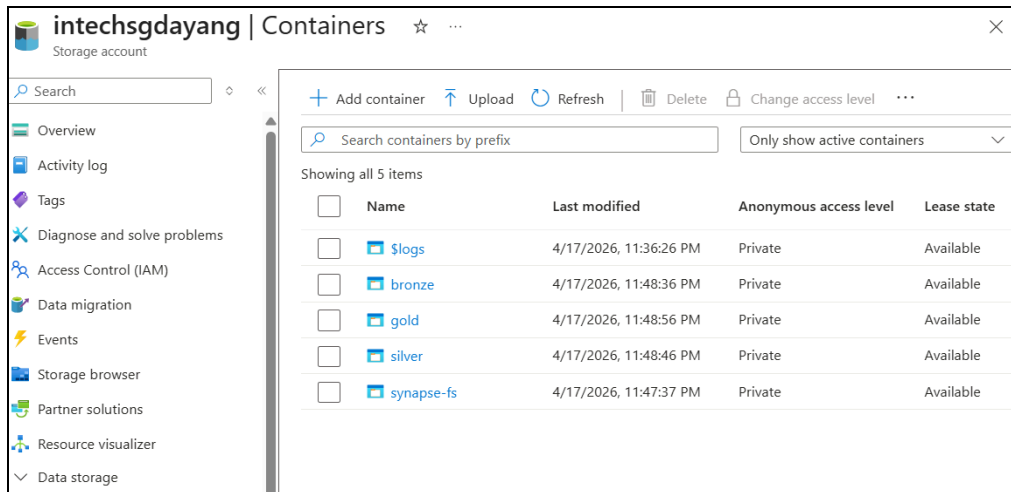


Figure 2.4: Created Bronze Layer in Storage Account

The data used in this project consists of customer records and sales transactions sourced from a sample AdventureWorks-style SQL database. Once the Copy Data activity runs successfully, the raw data is confirmed in the storage account as shown in Figure 2.5 and the pipeline is ready to trigger the next stage.

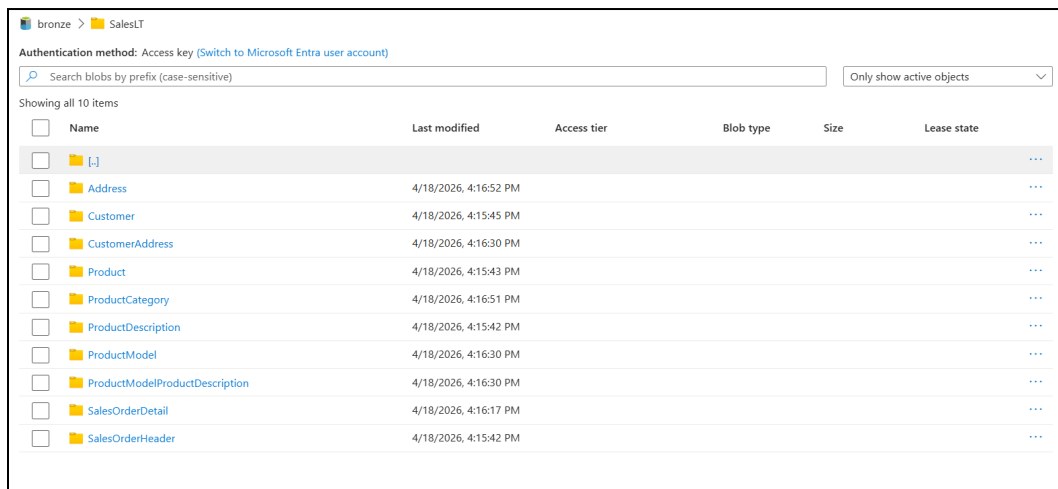


Figure 2.5: Raw Data in Bronze Layer Folder

After the raw data has been ingested into the Bronze layer, the next step is to process and transform it into a format that is ready for analytics. This is handled by Azure Databricks, which is a cloud-based Apache Spark environment that allows large-scale data transformations to be written in Python or SQL through interactive notebooks.

The processing follows a two-stage approach. Before any transformation begins, a Databricks cluster must first be created and started (running) as shown in Figure 2.6, as it provides the computational resources required to execute the notebooks and process the data. First, Figure 2.7 shows when the raw Bronze data is read into Databricks. Figure 2.8 and Figure

2.9 shows when the data is cleaned before and after it is written into the Silver Layer. Then, the Silver data is further refined and structured for reporting, and the result is written to the Gold layer. This medallion architecture pattern ensures that data quality improves at each step and that the original raw data is always preserved in case re-processing is needed.

The Silver layer transformations include:

- Converting date columns from string format to proper DateTime types as shown.
- Standardising column naming conventions to use consistent lowercase with underscores.

The Gold layer further aggregates and shapes the data specifically for the reporting use case, such as joining the sales and customer tables together so that gender and product category information are available in a single flat table for Power BI.

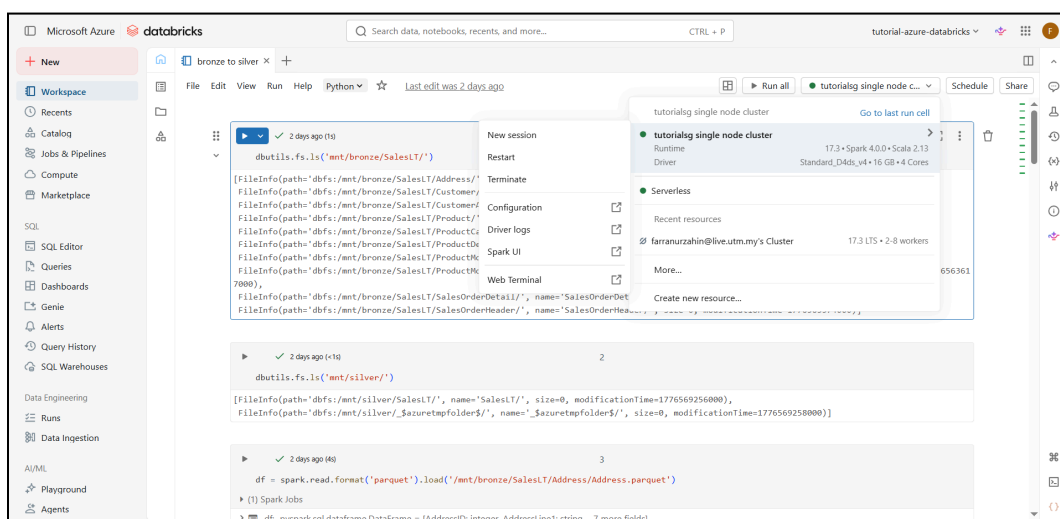


Figure 2.6: Run Databricks Cluster

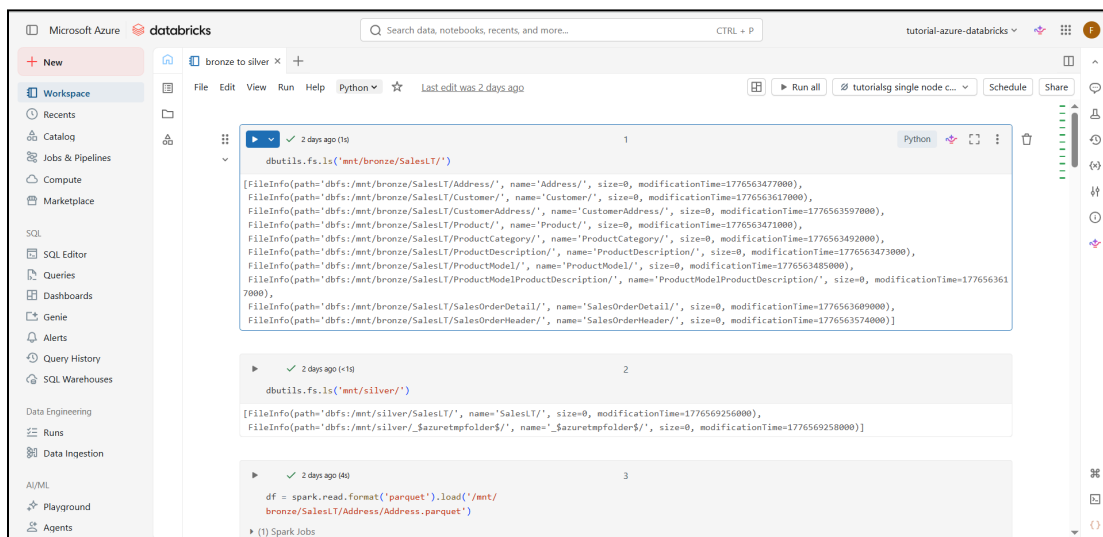


Figure 2.7: Data Transformation from Bronze to Silver Layer

Microsoft Azure databricks

Search data, notebooks, recents, and more...

tutorial-azure-databricks

File Edit View Run Help Python Last edit was 2 days ago

Run all tutorial:alg single node c...

display(df)

(1) Spark Jobs

i2	City	StateProvince	CountryRegion	PostalCode	rowguid	ModifiedDate
1	Bothell	Washington	United States	98011	268af621-76d7-4c78-9441-144d139821a	2006-07-01T00:00:00.000+00:...
2	Bothell	Washington	United States	98011	981b3303-aca2-49c7-9a96-fb670785b269	2007-04-01T00:00:00.000+00:...
3	Dallas	Texas	United States	75201	c8df3bd9-48f0-4654-a8dd-14a67a843c6	2006-09-01
4	Phoenix	Arizona	United States	85004	12ae5ee1-fc3e-4e6b-9b92-3b970b169774	2005-09-01
5	Montreal	Quebec	Canada	HTY 2H5	84a95f62-3ae8-4e7e-bbd5-5a6f00cd982d	2006-08-01
6	Bellevue	Washington	United States	98004	7bcc442-2268-46cc-8472-14c44c14e98c	2006-09-01
7	Renton	Washington	United States	98055	52410da4-2778-4b1d-a599-95746625cedf	2006-08-01
8	Burnaby	British Columbia	Canada	V5A 3A6	53572f25-9133-4a8b-a065-102f35416ee	2006-09-01
9	Toronto	Ontario	Canada	M4B 1V7	801a1dfc-5125-486b-a84-cbd2ec57ca4	2005-08-01
10	Toronto	Ontario	Canada	M4B 1V7	88cee379-dbb8-433b-b84e-a35e09435500	2006-08-01
11	Toronto	Ontario	Canada	M4B 1V6	2df6d0ad-0926-4f34-a450-9b1083150cbf	2007-08-01
12	Chalk River	Ontario	Canada	K0J 1J0	8b5a7729-cb75-4303-a607-79793b4d94f	2005-08-01
13	Quebec	Quebec	Canada	G1R	5f3c345a-6475-41d5-b17b-dbb827733bb1	2006-09-01
14	Calgary	Alberta	Canada	T2P 2G8	496441e-6f90-46d9-8dbb-9db150ef7ec	2006-12-01

450 rows | 7.46s runtime

Refreshed 2 days ago

Figure 2.8: ‘ModifiedDate’ Before Data Cleaning

Microsoft Azure databricks

Search data, notebooks, recents, and more...

tutorial-azure-databricks

File Edit View Run Help Python Last edit was 2 days ago

Run all tutorial:alg single node c...

display(df)

(1) Spark Jobs

ressLine2	City	StateProvince	CountryRegion	PostalCode	rowguid	ModifiedDate
1	Bothell	Washington	United States	98011	268af621-76d7-4c78-9441-144d139821a	2006-07-01
2	Bothell	Washington	United States	98011	981b3303-aca2-49c7-9a96-fb670785b269	2007-04-01
3	Dallas	Texas	United States	75201	c8df3bd9-48f0-4654-a8dd-14a67a843c6	2006-09-01
4	Phoenix	Arizona	United States	85004	12ae5ee1-fc3e-4e6b-9b92-3b970b169774	2005-09-01
5	Montreal	Quebec	Canada	HTY 2H5	84a95f62-3ae8-4e7e-bbd5-5a6f00cd982d	2006-08-01
6	Bellevue	Washington	United States	98004	7bcc442-2268-46cc-8472-14c44c14e98c	2006-09-01
7	Renton	Washington	United States	98055	52410da4-2778-4b1d-a599-95746625cedf	2006-08-01
8	Burnaby	British Columbia	Canada	V5A 3A6	53572f25-9133-4a8b-a065-102f35416ee	2006-09-01
9	Toronto	Ontario	Canada	M4B 1V7	801a1dfc-5125-486b-a84-cbd2ec57ca4	2005-08-01
10	Toronto	Ontario	Canada	M4B 1V7	88cee379-dbb8-433b-b84e-a35e09435500	2006-08-01
11	Toronto	Ontario	Canada	M4B 1V6	2df6d0ad-0926-4f34-a450-9b1083150cbf	2007-08-01
12	Chalk River	Ontario	Canada	K0J 1J0	8b5a7729-cb75-4303-a607-79793b4d94f	2005-08-01
13	Quebec	Quebec	Canada	G1R	5f3c345a-6475-41d5-b17b-dbb827733bb1	2006-09-01
14	Calgary	Alberta	Canada	T2P 2G8	496441e-6f90-46d9-8dbb-9db150ef7ec	2006-12-01

450 rows | 1.08s runtime

Refreshed 2 days ago

Figure 2.9: ‘ModifiedDate’ After Data Cleaning

Data cleaning is necessary in any real-world pipeline because source systems rarely produce perfectly structured data. Missing values, inconsistent formats, and duplicate records can all lead to inaccurate dashboard metrics if not addressed before reporting. The transformations applied here ensure that the Power BI dashboard reflects a clean and trustworthy dataset.

3.0 Technology Use

The pipeline uses six Azure services, each handling a specific part of the workflow:

1. Azure Data Factory (ADF) - ADF is the orchestration engine of the pipeline. It controls the sequence of all stages, from copying data from the SQL source to triggering the Databricks notebooks and can be scheduled to run automatically. All connections to other services are managed through linked services inside ADF.
2. Azure Data Lake Gen2 - The storage layer for all three data tiers. Bronze, Silver, and Gold data are stored as separate folders within the same storage account. Using Parquet format allows Databricks and Synapse to read the data efficiently.
3. Azure Databricks - Handles all transformation logic using PySpark notebooks. Databricks reads from the Bronze layer, applies cleaning steps (fixing date formats, removing nulls, standardising column names) and writes the results to Silver. It then joins and structures the data for the Gold layer. These notebooks are called directly from the ADF pipeline so transformation runs as part of the automated workflow.
4. Azure Key Vault - Figure 3.1 shows where Azure Key Vault stores sensitive credentials such as the storage account access key and connection strings. Instead of hardcoding these values into notebooks or pipeline configs, Databricks and ADF retrieve them from Key Vault at runtime using a service principal. This keeps credentials out of the codebase entirely.

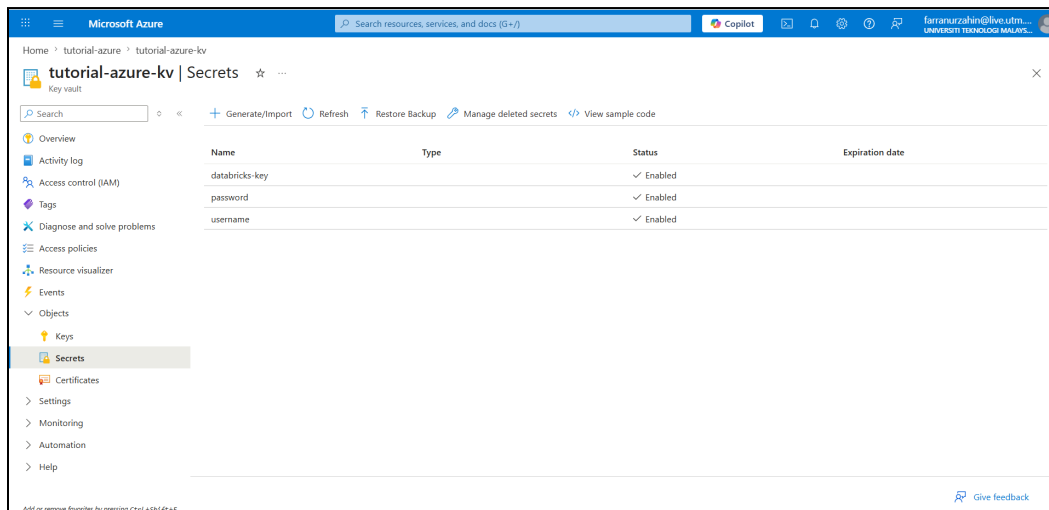


Figure 3.1: Stored Secret in Azure Key Vault

5. Azure Synapse Analytics - Exposes the Gold layer data for SQL querying through serverless SQL views. These views point directly to the Parquet files in the Data Lake without duplicating data. Synapse serves as the connection point between processed data and Power BI.
6. Microsoft Power BI - Connects to Synapse to retrieve the final dataset and presents it as an interactive dashboard. Users can filter and explore the data without needing technical knowledge of the underlying pipeline.

7. Azure Key Vault - Secures sensitive credentials used across services. Databricks retrieves the Data Lake access key from Key Vault at runtime using a service principal registered in Azure Active Directory (Entra ID), which avoids hardcoding secrets. Synapse connects to the same Data Lake through a linked service and Power BI connects to Synapse through its native Azure connector.

4.0 Data Pipeline Architecture

The full pipeline follows a layered flow from source to dashboard:

- On-Premises SQL Server - source of customer and sales transaction records.
- Azure Data Factory - orchestrates the pipeline; copies all tables from SQL to the Bronze layer via a ForEach loop.
- Data Lake Gen2 (Bronze Layer) - raw, unmodified copy of the source data stored as Parquet files.
- Azure Databricks (Silver Layer) - cleaned and formatted data after null removal, date fixes, and column standardisation.
- Azure Databricks (Gold Layer) - analytics-ready data with tables joined and structured for reporting.
- Azure Synapse Analytics - Gold data exposed via serverless SQL views; acts as the data source for Power BI.
- Microsoft Power BI - interactive dashboard visualizing the final processed data.

As shown in Figure 4.1, each layer has a clear purpose and data quality improves at every stage. Because ADF triggers the Databricks notebooks as part of the same pipeline, the entire flow from ingestion to the Gold layer runs automatically in sequence. Synapse and Power BI then consume the final output without any manual steps required.

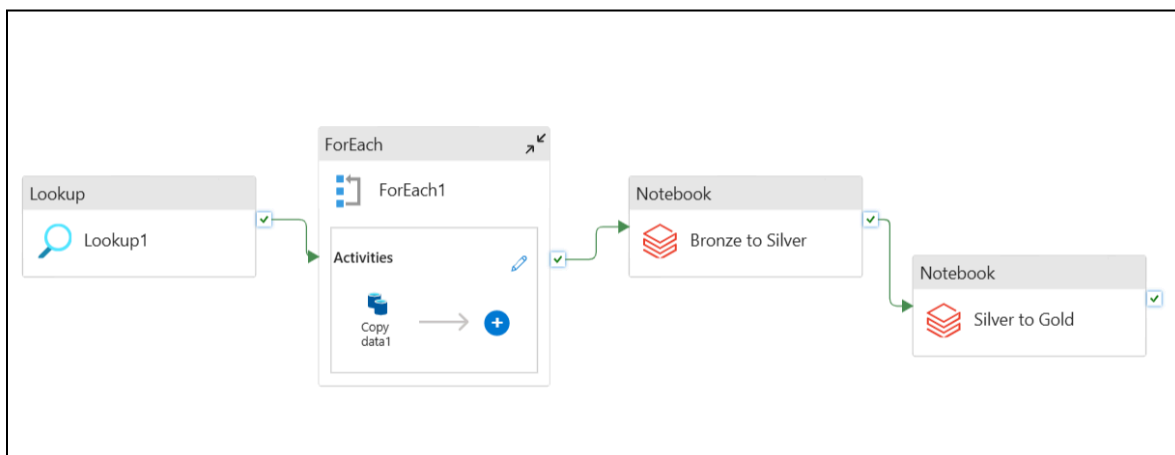


Figure 4.1 : Overview of Pipeline Architecture

5.0 MS Power BI Dashboard

Power BI connects to the Synapse serverless SQL endpoint and loads the views as tables. Relationships between the sales, customer, and product tables were defined in the model view so that filters on one visual carry through to all others on the page.

The dashboard in Figure 5.1, is built around two key business questions which are how total sales are split across a number of products, and what the title distribution of customers looks like. The visuals included are:

- Donut chart - title (Mr., Ms., Sr. vs Sra.).
- Card visuals - total sales and total number of products.
- Slicers - interactive filters for title and product name.

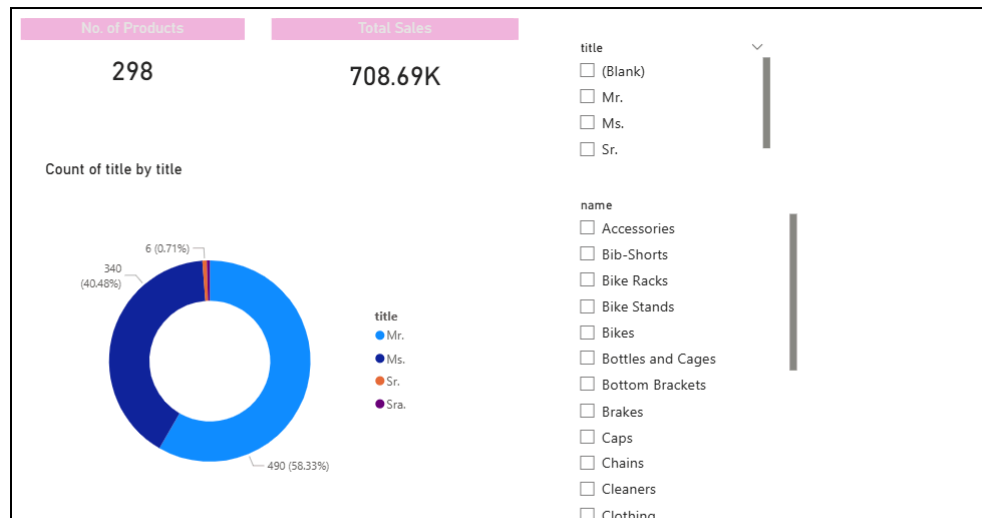


Figure 5.1 : PowerBI Dashboard

As shown in Figure 5.2 the dashboard was published to the Power BI service and configured to refresh automatically whenever the pipeline runs, so stakeholders always see up-to-date data without any manual action.

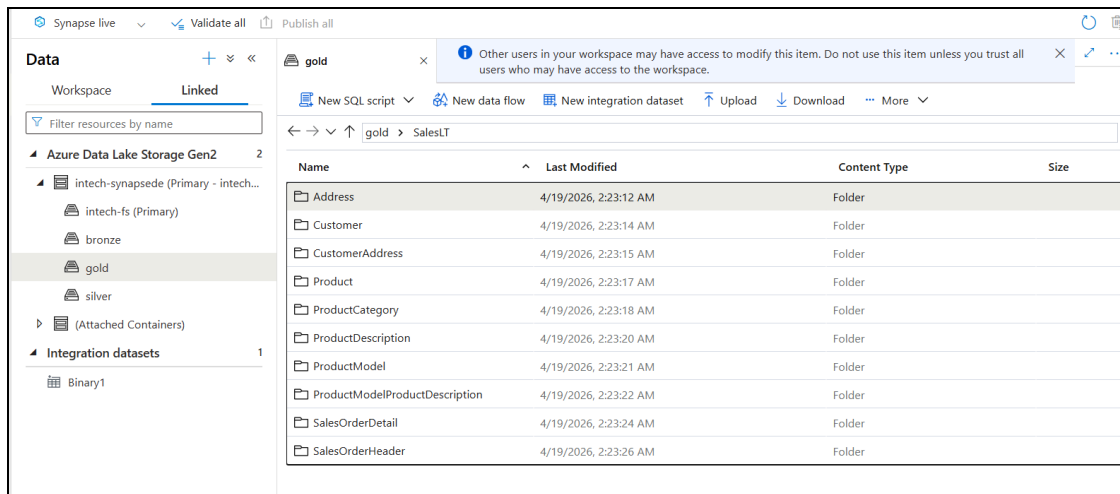


Figure 5.2 : Power BI dataset connection to Synapse

6.0 REFLECTION

Dayang

Throughout this project, several challenges were encountered, especially during the configuration and integration of different Azure services. There were issues related to configuration settings and coding, which required careful troubleshooting. To overcome this, AI tools were helpful in providing explanations and suggestions to identify and fix errors more efficiently. This experience helped improve problem-solving skills, particularly in debugging and understanding how different components in a data pipeline interact with each other. Overall, the project provided valuable hands-on experience in using Microsoft tools and building a complete data engineering workflow.

Farra

Following the tutorial from start to end helped me understand how each Azure service works together. At the beginning, setting up data ingestion in Azure Data Factory was a bit confusing and required careful configuration. In Databricks, I learned to create and run clusters, connect to Data Lake, and transform data through the Bronze, Silver, and Gold layers. Some steps needed a few tries before working correctly. In the end, using Synapse Analytics and Power BI showed how everything connects into a complete pipeline. Overall, this tutorial helped me understand the full data workflow and the importance of doing each step correctly.

Safiya

Following this tutorial gave me a solid understanding of how to build an end-to-end data pipeline on Microsoft Azure, starting from an on-premises SQL Server database all the way to a Power BI dashboard. The most challenging part was configuring the connections between the different Azure services, especially getting the file paths right between the Data Lake layers and Synapse. Overall, this project gave me practical hands-on experience with tools like Data Factory, Databricks, and Synapse that are commonly used in real data engineering work today.

